ClipSAT : Math for AI & Data Science

Unit 5.01: Distributions & Maximum Likelihood

Name: _____

Q1. A feature in an AI vision model follows a normal distribution X sim mathcal{N}5, 4 . What is the mean mu and variance sigma^2 of this feature?

[ ] A: mu = 5, ; sigma^2 = 4
[ ] B: mu = 4, ; sigma^2 = 5
[ ] C: mu = 5, ; sigma^2 = 2
[ ] D: mu = 2, ; sigma^2 = 4

Q2. A feature in an AI vision model follows a normal distribution X sim mathcal{N}10, 9 . What is the mean mu and variance sigma^2 of this feature?

[ ] A: mu = 10, ; sigma^2 = 3
[ ] B: mu = 9, ; sigma^2 = 10
[ ] C: mu = 10, ; sigma^2 = 9
[ ] D: mu = 3, ; sigma^2 = 9

Q3. A feature in an AI vision model follows a normal distribution X sim mathcal{N}15, 16 . What is the mean mu and variance sigma^2 of this feature?

[ ] A: mu = 15, ; sigma^2 = 16
[ ] B: mu = 16, ; sigma^2 = 15
[ ] C: mu = 15, ; sigma^2 = 4
[ ] D: mu = 4, ; sigma^2 = 16

Q4. A feature in an AI vision model follows a normal distribution X sim mathcal{N}20, 25 . What is the mean mu and variance sigma^2 of this feature?

[ ] A: mu = 20, ; sigma^2 = 25
[ ] B: mu = 25, ; sigma^2 = 20
[ ] C: mu = 20, ; sigma^2 = 5
[ ] D: mu = 5, ; sigma^2 = 25

Q5. A feature in an AI vision model follows a normal distribution X sim mathcal{N}25, 36 . What is the mean mu and variance sigma^2 of this feature?

[ ] A: mu = 25, ; sigma^2 = 36
[ ] B: mu = 36, ; sigma^2 = 25
[ ] C: mu = 25, ; sigma^2 = 6
[ ] D: mu = 6, ; sigma^2 = 36

Q6. A feature in an AI vision model follows a normal distribution X sim mathcal{N}30, 49 . What is the mean mu and variance sigma^2 of this feature?

[ ] A: mu = 30, ; sigma^2 = 49
[ ] B: mu = 49, ; sigma^2 = 30
[ ] C: mu = 30, ; sigma^2 = 7
[ ] D: mu = 7, ; sigma^2 = 49

Q7. Given n = 4 independent observations from a normal distribution with sum sum_{i=1}^{4} x_i = 12 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

[ ] A: hat{mu}_{text{MLE}} = 3
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 12
[ ] D: hat{mu}_{text{MLE}} = 1.5

Q8. Given n = 8 independent observations from a normal distribution with sum sum_{i=1}^{8} x_i = 24 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

[ ] A: hat{mu}_{text{MLE}} = 3

[ ] B: hat{mu}_{text{MLE}} = 6

[ ] C: hat{mu}_{text{MLE}} = 24

[ ] D: hat{mu}_{text{MLE}} = 1.5